Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/29/2006
PI Name:	Munir, Zuhair A Ph.D.		
Project Title:	Electric Field Effects in Self-Propagating High-Temperature Combustion Synthesis under Microgravity Conditions		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Surface combustion, shs, fire safety and smoldering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	95616	**Congressional District:**	3
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	99-HEDS-04
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No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Shaw, Benjamin, D. Ph.D. (Department of Mechanical & Aeronautical Engineering, University of California, Davis)		
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	The research will investigate the effect of an electric field on the dynamics of self-propagating high-temperature synthesis (SHS) waves and the effect of the field on phase formation and structure evolution in SHS. The planned work includes both terrestrial and reduced gravity experiments. The latter will be carried out in parabolic flights and in drop towers. The parabolic flight experiments will use existing facilities developed by Guigne International, Ltd. in NASA's KC-135 aircraft. Modifications were done to accommodate the requirements for application of an electric field application as well as instrumentation for temperature measurements and imaging. Drop tower experiments will be carried out in the NASA Glenn Research Center 2.2 s Drop Tower, and additional drop tower experiments will be carried out in Japan by one of the collaborating scientists. The research will employ systems that produce varied amounts of liquid. For example, the systems Si + C or Ti + Al allow investigation of the case where one reactant melts and the other does not, the system Ta + C allows investigation of a system where neither the reactants not the products melt, and the system FeAl allows investigation of the case where both reactants and products melt. These systems are expected to		

Task Description:	respond to the gravity level in different ways, e.g., through liquid transport effects. It is also planned that quenching experiments will be performed. This will be accomplished by turning off the electric field during the combustion process. The removal of the field will cause the deflagration wave to quench under certain conditions, essentially freezing the structure of the wave. Wave structure will be analyzed using XRD, SEM and EPMA. We plan to quantify heat losses from samples using several techniques. We will also pursue computational and asymptotic modeling efforts. The computational efforts involve extending previous numerical models to account for phenomena such as melting, capillary flows, buoyant flows and three spatial dimensions. Asymptotic studies will be concerned with determining influences of electric fields on stability and extinction of field-assisted SHS waves, taking into account capillary transport of liquids as well as local variations in surface tension and contact angles from thermal or electrical effects. This research program is focused on studying self-propagating high-temperature synthesis (SHS) under reduced gravity conditions. The broad objective of this research program is to understand the role of an electric field in SHS reactions under conditions where gravity-related effects are suppressed. The research will allow increased understanding of fundamental aspects of field-assisted SHS processes as well as synthesis of materials that cannot be formed in normal gravity. For example, the electric field has been shown to influence the nature of the product, the composition of phases in a composite, and the homogeneity (elemental distribution) in solid solutions. Furthermore, the field can influence other important SHS-related phenomena including capillary flow, mass-transport in porous media, and Marangoni flows. These phenomena are influenced by gravity in conventional SHS processes (i.e., without electric fields). As a result the influence of the field on SHS under reduced gravity is expected to be different than under normal gravity. It is also known that heat loss rates from samples, which can depend significantly on gravity, can influence final products in SHS. This research will provide new information on the effects of gravity on self-propagating high-temperature combustion synthesis with applied electric fields. The research will increase understanding of the effects of buoyancy, segregation, percolation and may potentially lead to new materials that cannot be formed under normal gravity conditions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The first flight campaign (KC-135) for the Microgravity Field Activated Combustion Synthesis (MFACS) experiments was completed with success on November 2003 at Johnson Space Center, Houston.After analysis it was found that the experimental temperatures and velocities in the Ti + Al system reached higher values under 1-g conditions compared to the 0-g conditions. Reaction temperatures and wave velocities increased in both cases with increasing currents applied, as expected. For this system it was found that the lowest current applied for the formation of stable combustion wave (threshold current) was higher during reduced gravity conditions, possibly affected by the decrease in the preheating of the sample due to the reduction of gravity driven convection effects. Average velocities were measured for the completed reactions in both 1-g and 0-g at 250A. It was found that the experiments in 1-g (1.92 ± 0.17 mm/s; ± standard deviation, s) had faster average velocities than the 0-g experiments (1.67 ± 0.06 mm/s; ± s).The Ti + 3 Al system was investigated to be able to understand the effects of different amounts of liquid involved in the reaction, with 3 times the aluminum compared to the Ti + Al system. Up until about 325A (threshold), terrestrial experiments displayed temperatures in excess of 375°C compared to reduced gravity experiments. A switch in this behavior was observed where the reduced gravity experiments showed a steep temperature increase up to about 600°C at 325A applied. After that, With increasing applied currents, the reduced gravity experiments had higher reaction temperatures up to highest current used in this study (550A). Microstructural evolution of the Ti + Al and Ti + 3Al systems have been investigated using SEM analysis and reaction sequences and zones have been identified in the quenched samples in both 1-g and 0-g.After ongoing analysis of the terrestrial and reduced gravity experiments it was hypothesized that increased percentage of the liquid phase compared to the solid phase would attenuate the effects of gravity or other phenomena at the absence of gravity. Problems with sample integrity and retaining constant contact with the electrodes simply limits the implementation of increased liquid phase. To avoid these factors, smaller samples that can contain up to 100% liquid were inserted into the center of a structurally stable green pellets. The exterior pellet is considered a chemical oven because it provides heat to the smaller insert by means of a combustion wave assisted by the electric current. Inserts composed of a heavy metal (W, Ta, TaC and Ti) and a low-density liquid (Al) investigated settling effects, which were mixed and pressed to full density. Percolation effects were studied by using inserts that consisted of loose powders of a metal (Ni and Ti) and liquid (Al). Furthermore, in order to isolate the electric current effects in microgravity, the chemical ovens contained different amounts of diluents, which allowed us to vary the current and retain very similar temperature profiles.The second flight campaign for the (MFACS) experiments was completed with success on August 2004 at Johnson Space Center, Houston. COSYM was returned to UC Davis for further terrestrial experiments. Initial results, indicated that experiments in 1-g showed settling of the heavier unreacted particles in the liquid Al within the short duration of the experiment (<25 seconds). In addition, throughout the W + Al inserts, intermetallic precipitates were formed with a high aspect ratio. In 0-g conditions no settling was found and heavier unreacted particles were distributed homogenously throughout the samples. Furthermore, precipitates with a lower aspect ratio were formed within the entire W + Al inserts.
Bibliography Type:	Description: (Last Updated: 02/27/2006)
Papers from Meeting Proceedings	Fredrick, D. M.; Unuvar, C.; Anselmi-Tamburini U.; Manerbino, A.; Guigne, J. Y.; Shaw, B. D.; Munir, Z. A. "Field activated combustion synthesis under zero gravity condition" Western State Section of the Combustion Institute Fall Meeting Proceedings and Presentation, University of California, Davis Mar-2004
Presentation	Munir, Z.A. and Shaw, B.D. "Observation on Gravity Dependence of Field Activated Combustion in Intermetallic Systems" Invited Presentation, The 2003 Japan-US Technology and Science Program (JUSTSAP) Workshop (on Microgravity), Honolulu, Hawaii Nov-2003
Presentation	Fredrick, D. M.; Unuvar, C.; Anselmi-Tamburini U.; Manerbino, A.; Guigne, J. Y.; Shaw, B. D.; Munir, Z. A. "Field activated combustion synthesis under zero gravity conditions" Western State Section of the Combustion Institute Fall Meeting Proceedings and Presentation University of California, Davis Mar-2004

Presentation	Sween, J. E. "Electric field and gravity effects on Combustion Synthesis reactions" 15th Annual Undergraduate Research Conference Presentation, University of California, Davis Apr-2004
Presentation	Shaw, B. D.; Unuvar, C.; Fredrick, D. M.; Anselmi-Tamburini U.; Manerbino, A.; Guigne, J. Y.; Munir, Z.A. "Electric Current Activated Combustion Synthesis And Chemical Ovens Under Terrestrial And Reduced Gravity Conditions" Conference-Workshop on Strategic Research to Enable NASA's Exploration Missions, Poster Presentation, Cleveland, Ohio USA Jun-2004
Presentation	Unuvar, C.; Fredrick, D. M.; Anselmi-Tamburini U.; Manerbino, A.; Guigne, J. Y.; Shaw, B. D.; Munir, Z.A. "Field activated combustion synthesis of Ti+Al and Ni+Al systems under zero gravity conditions" 56th Pacific Coast Regional and Basic Science Division Fall Meeting Presentation, Seattle, Washington, USA. Sep-2004